Unemployment and Inflation

ECON 101 - Macroeconomics

Muhebullah Karimzada

Winter 2026

Chapter Outline

- **5.1** Measuring the Unemployment Rate and the Labour Force Participation Rate
- **5.2** Types of Unemployment
- **5.3** Explaining Unemployment
- **5.4** Measuring Inflation
- **5.5** Using Price Indexes to Adjust for the Effects of Inflation
- **5.6** Real versus Nominal Interest Rates
- **5.7** Does Inflation Impose Costs on the Economy?

Measuring Unemployment and Inflation

- Previous chapter: how to measure **total output** (GDP).
- This chapter: how to measure
  - **Unemployment** in the labour market, and
  - **Inflation** in the goods and services market.
- Both are central for:
  - macroeconomic policy,
  - evaluating economic performance,
  - everyday decisions of households and firms.

5.1 Learning Objective

Goal

Understand how unemployment, labour force participation, and employment rates are measured.

- Canada is a large economy – millions of people, changing jobs all the time.
- We cannot track every person individually in real time.
- Instead, Statistics Canada uses **survey data** to estimate:
  - employment,
  - unemployment,
  - labour force participation,
  - employment-population ratio.

The Labour Force and Unemployment

- **Labour force:** sum of employed and unemployed workers in the economy.
- **Unemployment rate:** percentage of the labour force that is unemployed:

$$\text{Unemployment rate} = \frac{\text{Number of unemployed}}{\text{Labour force}} \times 100.$$

- These are monthly headline numbers often reported in the news.

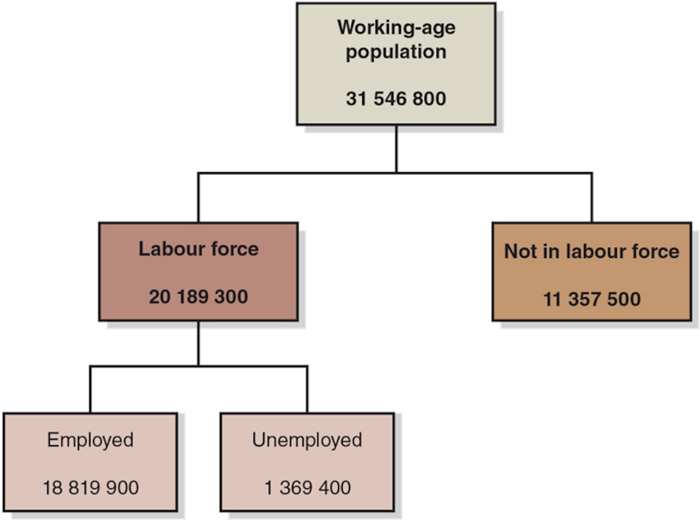
The Labour Force Survey (LFS)

- Each month, Statistics Canada conducts the **Labour Force Survey (LFS)**.
- Roughly **56,000 households** are selected to be representative.
- Members of **working age** (15 years and older) are:
  - asked about employment status, and
  - asked about recent **job search** activities.
- Based on the answers, each person is classified as:
  - **Employed**
  - **Unemployed**
  - **Not in the labour force**

Definitions: Employed, Unemployed, Not in the Labour Force

- **Employed:**
  - Did paid work, or
  - Did unpaid work for a family business, or
  - Worked for themselves, or
  - Were temporarily away from their jobs (vacation, illness, strike).
- **Unemployed:**
  - Not currently at work,
  - *Available* for work,
  - Actively looked for work in the previous four weeks.
- **Not in the labour force:**
  - Neither employed nor unemployed (students not seeking work, retirees, etc.).

Figure 5.1 – Employment Status of the Working-Age Population



Labour Force Participation Rate

- **Labour force participation rate:**

$$\text{LFPR} = \frac{\text{Labour force}}{\text{Working-age population}} \times 100.$$

- Measures what fraction of working-age people are **actively** in the labour market.
- Helps us distinguish:
  - low unemployment because people have jobs,
  - vs. low unemployment because many have given up looking for work.

Employment-Population Ratio

- **Employment-population ratio:**

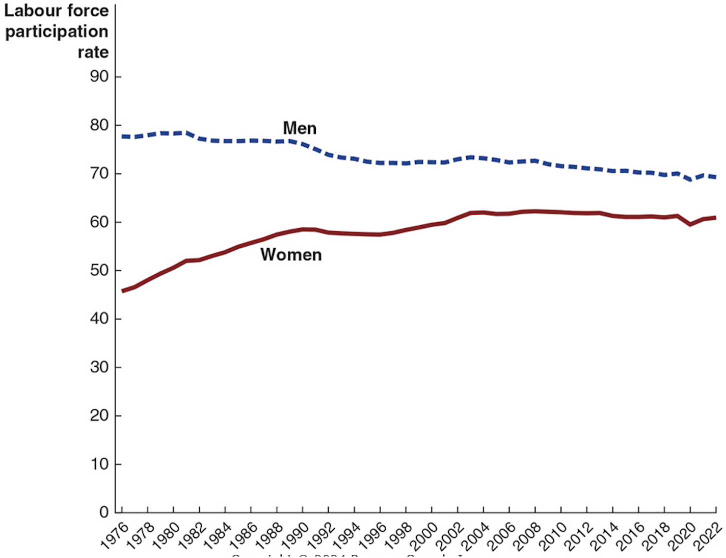
$$\text{Employment-population ratio} = \frac{\text{Number employed}}{\text{Working-age population}} \times 100.$$

- This tells us how many people **actually have jobs**, relative to the working-age population.
- Useful indicator of the **health of the labour market** during expansions and recessions.

Problems with Measuring the Unemployment Rate

- The measured unemployment rate is **not perfect**.
- It may understate true unemployment:
  - Distinguishing between **unemployed** and **not in the labour force** can be difficult.
  - **Discouraged workers**: people who have stopped looking for work but would take a job if offered one.
  - It does not measure intensity of employment — e.g., part-time vs. full-time, underemployment.
- It may overstate unemployment:
  - Some people may claim to be unemployed while not seriously looking, to keep benefits or avoid taxes.

Figure 5.2 – Trends in Labour Force Participation



Job Creation and Job Destruction

- The labour market is **dynamic**:
  - Firms constantly create and destroy jobs.
- Example (from the slide):
  - In January 2022, there were 890,500 more people employed than in January 2021.
  - But the number of jobs **created** was much larger, because many jobs were also **destroyed**.
- Month to month, some provinces can see large job losses even while the national economy is improving.

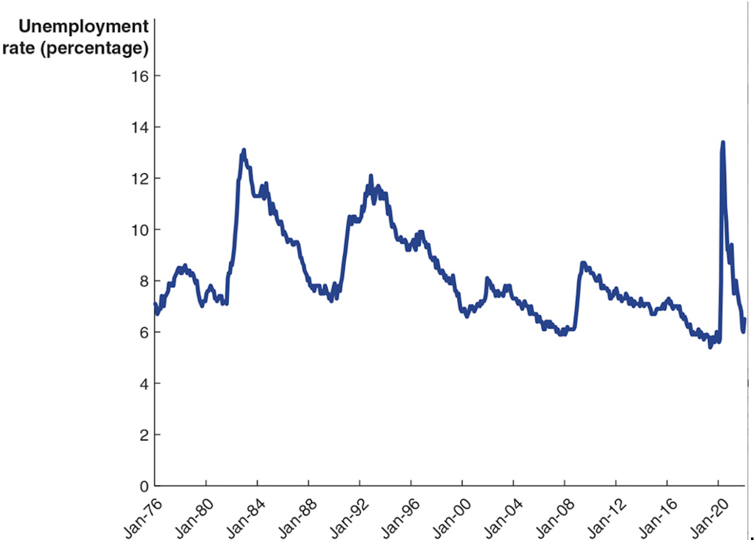
5.2 Learning Objective

Goal

Identify and understand the four types of unemployment.

- Unemployment is not a single phenomenon.
- To interpret unemployment correctly, we distinguish:
  - frictional,
  - structural,
  - cyclical,
  - seasonal unemployment.

Figure 5.3 – The Unemployment Rate in Canada



Four Types of Unemployment

- **Frictional unemployment**
- **Structural unemployment**
- **Cyclical unemployment**
- **Seasonal unemployment**
- Each type has different causes and different policy implications.

Frictional Unemployment

- **Frictional unemployment:**
  - Short-term unemployment that arises from the process of matching workers with jobs.
- Mainly due to **job search**:
  - People entering or re-entering the labour force.
  - People between jobs.
- Some frictional unemployment is unavoidable:
  - People need time to learn about opportunities and find good matches.
- In fact, some frictional unemployment can increase economic efficiency:
  - It allows workers to find jobs that better fit their skills and preferences.

Example: Frictional Unemployment

- A recent university graduate spends three months searching for a job in their field.
- They are counted as **unemployed**, but this is frictional unemployment.
- From the economy's perspective:
  - It is better for them to search a bit longer and find a suitable, productive job than to accept the first random job.

Structural Unemployment

- **Structural unemployment:**
  - Unemployment that arises from a persistent mismatch between the skills and attributes of workers and the requirements of jobs.
- Often associated with:
  - technological change,
  - changes in demand for certain industries,
  - regional shifts in economic activity.
- Structural unemployment tends to be **longer-term**:
  - Workers may need retraining or relocation.

Example: Structural Unemployment

- A manufacturing worker loses their job due to automation.
- Their skills do not easily transfer to growing sectors such as IT or health care.
- Without retraining, they may remain unemployed for a long period.

Cyclical Unemployment

- **Cyclical unemployment:**
  - Unemployment caused by a business cycle recession.
- When aggregate demand falls:
  - firms reduce output,
  - and lay off workers even if workers are skilled and jobs are otherwise a good match.
- Example:
  - In February 2009, Chrysler temporarily closed its assembly plant in Brampton, Ontario, due to low sales.
  - Laid-off employees experienced **cyclical unemployment**.

Seasonal Unemployment

- **Seasonal unemployment:**
  - Unemployment due to seasonal factors such as weather or calendar patterns.
- Examples:
  - Construction and tourism often slow in winter.
  - Agriculture, fisheries, and some retail sectors are highly seasonal.
- Seasonal unemployment can make the measured unemployment rate:
  - artificially high in winter,
  - artificially low in summer.
- To address this, Statistics Canada provides:
  - **seasonally adjusted** series (seasonal effects removed),
  - **unadjusted** series (seasonal effects included).

Full Employment and the Natural Rate of Unemployment

- Even in good times, there is always some unemployment.
- When all unemployment is due to frictional and structural factors, the economy is at **full employment**.
- **Natural rate of unemployment:**
  - The normal rate of unemployment, consisting of frictional and structural unemployment.
- Estimates for Canada suggest a natural rate around **5.6–6.7%** in recent years.

5.3 Learning Objective

Goal

Explain what determines the unemployment rate and how policy affects it.

- Governments often try to influence unemployment directly and indirectly.
- Some policies reduce frictional or structural unemployment.
- Other policies can unintentionally increase unemployment.

Government Policies and Unemployment

- Examples of policies that can **reduce unemployment**:
  - Retraining programs (e.g., Ontario's "Better Jobs Ontario") to reduce structural unemployment.
  - Hiring subsidies or job search assistance to reduce frictional unemployment.
- Examples of policies that might **increase unemployment**:
  - Employment insurance (if very generous),
  - Minimum wage laws that raise the cost of hiring,
  - Some labour market regulations.

Employment Insurance (EI)

- Suppose you lose your job. You can:
  - take a new low-paying job immediately, or
  - search for a better job that matches your skills.
- If **employment insurance** payments are available, you are more likely to choose the second option.
- In Canada, EI replaces about **55%** of earnings up to a weekly maximum (e.g., $595 per week in 2021).
- EI effects:
  - Allows more time for job search $\Rightarrow$ better matches, higher productivity.
  - But also tends to increase the average duration of unemployment.

Minimum Wage Laws

- Minimum wage laws are designed to help low-income workers.
- Raising the minimum wage increases the **cost of labour**.
- Firms may respond by:
  - hiring fewer workers,
  - substituting capital for labour,
  - reducing hours.
- In Canada, minimum wages vary by province:

Region	Example Minimum Wage (June 2022)
Nunavut	\$16.00/hour
New Brunswick	\$12.75/hour

- Studies suggest a 10% increase in the minimum wage might reduce teenage employment by about 2%.
- Overall effect on total unemployment is believed to be **relatively small**.

Labour Unions

- **Labour unions:** organizations of workers that bargain with employers for:
  - higher wages,
  - better working conditions.
- Unions can push wages above the competitive level for their members.
- In Canada:
  - about **30.9%** of workers belonged to a union by the end of 2021,
  - roughly **77.2%** of government employees are unionized,
  - only about **15.3%** of private-sector workers are unionized.
- Conclusion: unions likely have **limited impact** on overall unemployment in Canada.

Efficiency Wages

- **Efficiency wage:** a wage above the market-clearing level that firms pay to increase workers' productivity.
- Why pay more than necessary?
  - Higher wages can reduce worker turnover.
  - They can increase effort and reduce shirking.
  - They can attract higher-quality applicants.
- But paying an efficiency wage means:
  - firms hire fewer workers,
  - some workers who want jobs at that wage cannot find them.
- This contributes to **persistent unemployment**, even when the economy is not in recession.

5.4 Learning Objective

Goal

Define the price level and inflation rate, and understand how they are computed.

- We introduced the **price level** in the previous chapter using the GDP deflator.
- Now we look at more detailed measures used in practice:
  - the **Consumer Price Index (CPI)**,
  - the **Producer Price Index (PPI)**.

Price Level and Inflation Rate

- **Price level:** a measure of the average prices of goods and services in the economy.
- **Inflation rate:** the percentage increase in the price level from one year to the next:

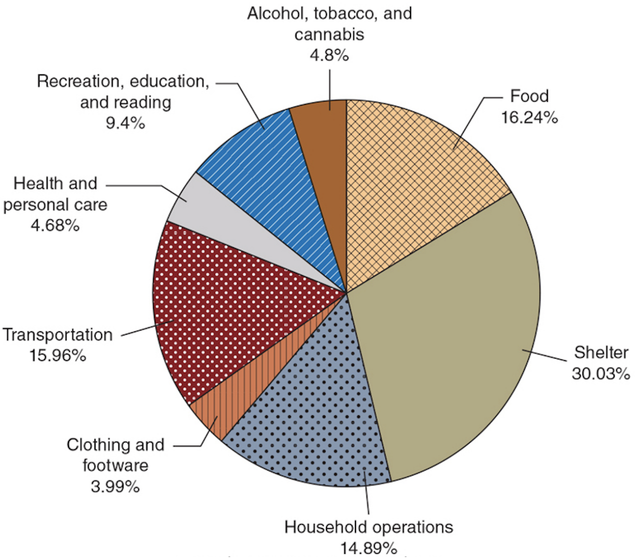
$$\text{Inflation rate} = \frac{\text{Price level}_t - \text{Price level}_{t-1}}{\text{Price level}_{t-1}} \times 100.$$

- In Chapter 4, we used the **GDP deflator** to measure the price level for all final goods and services.
- However, households and firms are often more interested in specific baskets of goods.

The Consumer Price Index (CPI)

- **CPI:** measures the average change over time in the prices paid by a typical household for the goods and services they purchase.
- It is based on a fixed **basket of goods and services**.
- Example categories:
  - food, shelter, transportation, clothing, recreation, etc.
- CPI is widely used to:
  - index wages,
  - adjust pensions and benefits,
  - compare the cost of living over time.

Figure 5.4 – The CPI Basket



Calculating the CPI

- To compute the CPI for a given year, we need:
  - A fixed **basket of goods and services** (quantities from a base year).
  - The cost of buying the basket in the **base year**.
  - The cost of buying the basket in the **current year**.
- Then:

$$\text{CPI}_t = \frac{\text{Cost of basket in year } t}{\text{Cost of basket in base year}} \times 100.$$

- By convention, $\text{CPI}_{\text{base year}} = 100$.

A Simple CPI Calculation (1 of 2)

- Suppose the CPI basket (from 2002) has:
  - 10 units of good A,
  - 5 units of good B,
  - 2 units of good C.
- We know prices for 2002, 2022, and 2023 from the slide table.
- **Cost of basket in 2002:**

$$\text{Cost}_{2002} = 10P_A^{2002} + 5P_B^{2002} + 2P_C^{2002}.$$

A Simple CPI Calculation (2 of 2)

- **Cost of basket in 2022:**

$$\text{Cost}_{2022} = 10P_A^{2022} + 5P_B^{2022} + 2P_C^{2022}.$$

- **CPI in 2022:**

$$\text{CPI}_{2022} = \frac{\text{Cost}_{2022}}{\text{Cost}_{2002}} \times 100.$$

- Similarly for 2023.
- **Inflation between 2022 and 2023:**

$$\text{Inflation} = \frac{\text{CPI}_{2023} - \text{CPI}_{2022}}{\text{CPI}_{2022}} \times 100.$$

- This inflation rate is often called **CPI inflation** or a **cost-of-living** measure.

Apply the Concept: Were Prices Really Cheaper in the Past?

- Example from the slides:
  - In 1915, a pair of cashmere socks cost \$0.35.
  - In 2021, a similar pair costs \$39.00.
- Nominally, socks are much more expensive today.
- But incomes are also much higher.
- Using the CPI to convert \$0.35 (1915) into 2021 dollars gives about \$8.13.
- Compared with the actual 2021 price of \$39, socks are more expensive in real terms, but relative to **average income**, households today can afford many more socks than in the past.

Is the CPI Accurate?

- The CPI is widely used, but it is not perfect.
- Four main biases:
  - **Substitution bias**
  - **Increase in quality bias**
  - **New product bias**
  - **Outlet bias**
- Together, these biases tend to make CPI inflation overstate the true increase in the cost of living (by about 0.5 to 1 percentage point per year).

CPI Biases in More Detail

- **Substitution bias:**
  - Basket is fixed; in reality, consumers substitute toward relatively cheaper goods.
- **Quality change bias:**
  - Part of the price increase reflects better quality (e.g., safer, more efficient cars).
- **New product bias:**
  - New goods (e.g., smartphones, tablets) enter the basket with a delay.
- **Outlet bias:**
  - Shift from traditional stores to discount outlets or online retailers is not fully captured by the fixed basket.

Producer Price Index (PPI)

- **PPI:** an average of the prices received by producers of goods and services at all stages of production.
- Uses a basket of:
  - raw materials,
  - intermediate goods,
  - goods sold by wholesalers.
- Changes in the PPI often signal future movements in consumer prices:
  - higher input costs today tend to lead to higher retail prices later.

5.5 Learning Objective

Goal

Use price indexes to adjust nominal values for inflation.

- Often we want to compare **purchasing power** across years.
- To do this, we convert nominal amounts into **real** amounts using a price index like the CPI.

Adjusting Nominal Values for Inflation

- Suppose your mother earned \$30,000 in 1993.
- \$30,000 in 1993 could buy more goods and services than \$30,000 in 2021.
- To express the 1993 salary in 2021 dollars:

$$\text{Value in 2021 dollars} = \text{Value in 1993 dollars} \times \frac{\text{CPI}_{2021}}{\text{CPI}_{1993}}.$$

- Using the actual CPI numbers from the slides, this yields about \$49,626 for 2021.

Practical Use: Indexing Wages and Benefits

- Unions and employers often negotiate **cost-of-living adjustments (COLAs)**.
- Government programs (pensions, social benefits) are often indexed to CPI.
- This helps protect real purchasing power when prices rise.

5.6 Learning Objective

Goal

Distinguish between nominal and real interest rates.

- Interest rates are crucial for:
  - savers and borrowers,
  - investment decisions,
  - monetary policy.

Inflation and Interest Rates

- When you lend money, the borrower pays you back with interest.
- If the **nominal interest rate** is 6%, a \$1,000 loan repaid in one year returns \$1,060.
- But prices may rise over that year, reducing the **real purchasing power** of \$1,060.
- To understand the true cost of borrowing and lending, we use the **real interest rate**.

Nominal vs. Real Interest Rate

- **Nominal interest rate:** the stated interest rate on a loan.
- **Real interest rate** adjusts for inflation:

$$\text{Real interest rate} \approx \text{Nominal interest rate} - \text{Inflation rate}.$$

- Example:
  - If nominal rate is 6% and inflation is 2%, the real rate is about 4%.
- For low to moderate inflation, this approximation is very accurate and widely used in macroeconomics.

5.7 Learning Objective

Goal

Discuss the problems inflation can cause, even when anticipated.

- At first glance, inflation might seem harmless if all prices and wages doubled together.
- But in reality, inflation creates **winners and losers** and can distort decisions.

Is Inflation a Problem?

- If all prices and wages rose at exactly the same rate:
  - relative prices would not change,
  - everyone's purchasing power would stay the same.
- In practice:
  - not all prices adjust at the same time or by the same amount,
  - some wages are fixed in long-term contracts,
  - some assets (like cash) lose value.
- So inflation affects:
  - the **distribution of income and wealth**,
  - the incentives to save, borrow, and invest.

Problems with Anticipated Inflation

- Even if inflation is **anticipated**, it can create costs:
  - **Redistribution of income**:
    - Some workers' wages adjust fully to inflation; others lag behind.
  - **Loss of purchasing power of cash**:
    - People holding currency see its real value fall.
  - **Menu costs**:
    - Firms incur costs of changing prices (new catalogues, re-tagging, updating systems).
  - **Bracket creep and taxes on nominal returns**:
    - Investors are taxed on nominal, not real, returns; inflation increases the tax burden even if real returns are unchanged.

Problems with Unanticipated Inflation

- **Unanticipated inflation** is especially damaging.
- It makes it hard to write long-term contracts.
- Borrowers and lenders are exposed to risk:
  - If inflation turns out higher than expected:
    - borrowers gain (repay in less valuable dollars),
    - lenders lose (receive lower real interest).
  - If inflation turns out lower than expected:
    - borrowers lose,
    - lenders gain.
- Unpredictable inflation discourages saving and long-term investment.

Example: High and Uncertain Inflation

- In periods of high and unstable inflation (e.g., late 1970s–early 1980s):
  - Mortgage interest rates climbed into double digits (18% or more).
  - Banks tried to protect themselves from high and uncertain inflation.
- Households who locked in at high nominal rates suffered later when inflation fell but their nominal rates did not.
- These experiences show why central banks place high value on **low and stable inflation**.

Summary

- The unemployment rate, labour force participation rate, and employment-population ratio summarize labour market conditions.
- Unemployment has four main types: frictional, structural, cyclical, and seasonal.
- Policies such as EI, minimum wage laws, unions, and efficiency wages affect unemployment in different ways.
- Inflation is measured using price indexes like the CPI, PPI, and GDP deflator.
- The CPI is a cost-of-living index but contains several biases.
- We use price indexes to convert nominal values into real values.
- Real interest rates matter for saving, borrowing, and investment decisions.
- Inflation, especially when unanticipated, can impose significant costs on the economy.

Thank you!